HOW TO RUN THE APPLICATION

A Data-Budget-Aware and Explanation-Stable Lightweight Deep Learning Framework for Plant Disease Detection

Step-by-step guide for running the Plant Disease Detection web application on any Windows laptop.

Quick start – one command does everything

Copy `PlantDiseaseApp.zip` into the **Downloads** folder. Open **PowerShell** (press the Windows key, type `powershell`, press Enter), paste this one line and press **Enter**:

```
cd ~\Downloads; if (!(Test-Path PlantDiseaseApp)) { Expand-Archive PlantDiseaseApp.zip . }; .\PlantDiseaseApp\run_app.bat
```

This single command:

- extracts the zip (only the first time, so your accounts and history are never overwritten);
- finds Python 3.10–3.12, and **installs Python 3.12 automatically** with winget if it is missing;
- creates a virtual environment and installs PyTorch, Torchvision, Flask, Pillow and Matplotlib (first time only, 5–10 minutes, needs internet);
- starts the app and opens **http://127.0.0.1:5000** in the browser.

Next time, run the **same command** again – it starts in a few seconds without internet. The sections below explain each step in detail.

1. What you need

Item	Requirement
Computer	Windows 10 or Windows 11 laptop (no GPU needed)
Memory / disk	4 GB RAM or more, about 1.5 GB free disk space
Python	Python **3.10, 3.11 or 3.12** (3.12 recommended)
Internet	Needed **only for the first run**, to download the packages (~250 MB)
Browser	Google Chrome, Microsoft Edge or Firefox
Project file	`PlantDiseaseApp.zip` (about 6 MB)

2. Install Python (one time only)

`run_app.bat` installs Python automatically when it is missing. Install it by hand only if the automatic installation fails:

1. Open **https://www.python.org/downloads/** and download **Python 3.12** for Windows.
2. Run the installer. On the first screen **tick "Add python.exe to PATH"**, then click **Install Now**.
3. To check, open **Command Prompt** (press Windows key, type `cmd`, press Enter) and type the command below. It should print a version such as Python 3.12.x.

```
python --version
```

3. Copy and extract the project

1. Copy `PlantDiseaseApp.zip` to the laptop (pen drive, Google Drive, e-mail, etc.).
2. Right-click the zip file → **Extract All…** → **Extract**.
3. Open the extracted **PlantDiseaseApp** folder. It contains:

File / folder	Purpose
`run_app.bat`	One-click setup and start
`app.py`	Flask web application
`src`	Grad-CAM and prediction code
`model_bundle`	Trained MobileNetV3-Small model, class list, settings
`templates`, `static`	Web pages, style sheet, uploaded images
`instance`	SQLite database with user accounts and history
`requirements.txt`	List of Python packages

4. Run the application (easy method)

1. Double-click **run_app.bat**. A black Command Prompt window opens.
2. If Windows shows "Windows protected your PC", click **More info** → **Run anyway**.
3. **First run only:** the window shows the steps **[1/3]** to **[3/3]**. It installs Python if needed, creates a virtual environment in `%USERPROFILE%\.plantapp_venv` and installs PyTorch, Torchvision, Flask, Pillow and Matplotlib. This takes about **5–10 minutes**. Keep the internet on and do not close the window.
4. When you see **"Running on http://127.0.0.1:5000"**, the browser opens the app automatically. If the page does not load, wait a few seconds and press **F5**.
5. **Next runs:** double-click **run_app.bat** again. It starts in a few seconds, without internet.

5. Run the application (manual method)

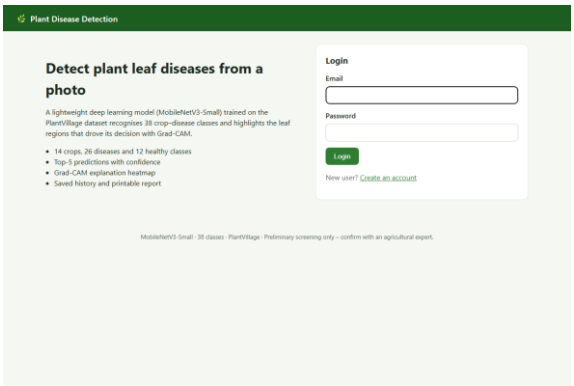
Use this if the batch file does not work. Open Command Prompt in the **PlantDiseaseApp** folder (type `cmd` in the folder's address bar and press Enter) and run:

```
python -m venv %USERPROFILE%\.plantapp_venv
%USERPROFILE%\.plantapp_venv\Scripts\activate
pip install -r requirements.txt
python app.py
```

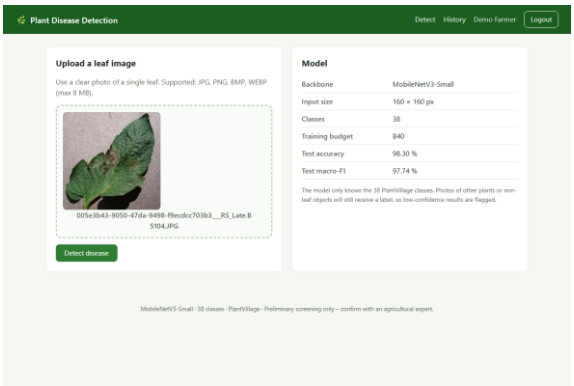
Then open **http://127.0.0.1:5000** in the browser. Next time, only the second and the last command are needed. The virtual environment is kept in a short path because some PyTorch file names are very long.

6. Using the application

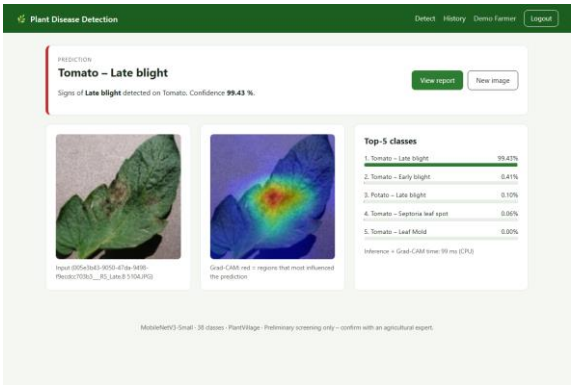
1. **Register** – on the login page click **Create an account**, enter name, e-mail and a password (at least 6 characters). Accounts created on the original laptop are already included, so you can also log in with them directly.
2. **Login** – enter the e-mail and password and click **Login**.
3. **Upload** – choose a leaf photo (JPG, PNG, BMP or WEBP) and click **Detect disease**.
4. **Result** – the page shows the crop, the disease, the confidence, the top predictions and the Grad-CAM heat map. A warning appears when confidence is below 50%.
5. **Report** – click **View report**, then **Print / Save PDF** to print or save it.
6. **History** – click **History** at the top to see all previous predictions of the logged-in user.
7. **Logout** – click Logout at the top right.



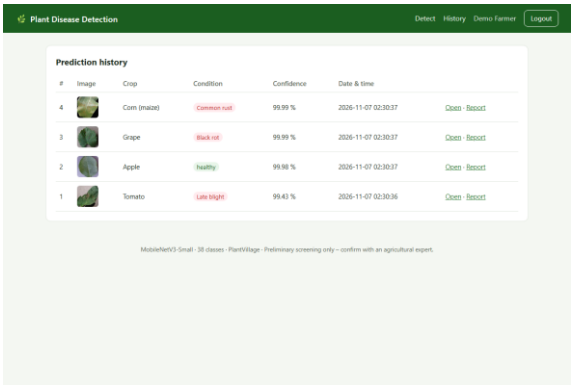
Login page



Upload a leaf image



Prediction with Grad-CAM



History page

7. Stopping the application

Press **Ctrl + C** in the black Command Prompt window, or simply close that window. The browser page stops working after the server is stopped.

8. Troubleshooting

Problem	Solution
"Python could not be installed automatically"	Install 64-bit Python 3.12 by hand (Section 2), tick **Add python.exe to PATH**, and run the command again.
Package installation fails	Check the internet connection and run the command again. The script removes the half-installed environment by itself.
Upgrade or reinstall all packages	Delete the folder `%USERPROFILE%\.plantapp_venv` and run the command again.
"No matching distribution found for torch"	The Python version is too new or 32-bit. Install **64-bit Python 3.12**.
Browser shows "This site can't be reached"	The server is still loading – wait a few seconds and press F5. Make sure the black window is still open.
"Address already in use" / port 5000 busy	Another copy of the app is running. Close the other black window, or use another port: in PowerShell run `$env:PORT=5001; .\PlantDiseaseApp\run_app.bat` and open http://127.0.0.1:5001.
"Invalid email or password"	Check the e-mail and password, or click **Register** to create a new account.
"Unsupported file type" or "too large"	Use a JPG, PNG, BMP or WEBP image smaller than 8 MB.

9. Running the tests (optional)

The automated tests are not part of the zip. In the full project folder run:

```
pip install pytest
python -m pytest tests -q
```

All 8 tests should pass.